

Department of Computer Science & Engineering

Lab Manual – 2301CS301 – Data Structure

B. Tech. Semester – III | Extra Practice Questions | Academic Year 2026-27

No.	Problem	Approach	Difficulty
Unit: II – Linked List			
1	LC: 2. Add Two Numbers	Two pointers + carry simulation	Med
2	LC: 21. Merge Two Sorted Lists	Merge with dummy node	Easy
3	LC: 23. Merge k Sorted Lists	Min-heap / divide & conquer	Hard
4	LC: 61. Rotate List	Find cycle length, two pointers	Med
5	LC: 83. Remove Duplicates from Sorted List	Single pass	Easy
6	LC: 82. Remove Duplicates from Sorted List II	Dummy node + skip duplicates	Med
7	LC: 86. Partition List	Two dummy pointers (< and >=)	Med
8	LC: 206. Reverse Linked List	Iterative pointer reversal	Easy
9	LC: 92. Reverse Linked List II	Reverse sub-list with 4 pointers	Med
10	LC: 141. Linked List Cycle	Floyd's cycle detection	Easy
11	LC: 142. Linked List Cycle II	Floyd's cycle detection (entry point)	Med
12	LC: 148. Sort List	Merge sort on linked list	Med
13	LC: 234. Palindrome Linked List	Fast/slow pointer + reverse half	Easy
14	LC: 2487. Remove Nodes From Linked List	Monotonic stack (decreasing)	Med
15	LC: 2296. Design a Text Editor	Doubly linked list + stack	Hard
Unit: I – 1D & 2D Arrays			
16	LC: 88. Merge Sorted Array	3 pointers (merge from end)	Easy
17	LC: 53. Maximum Subarray	Kadane's algo	Med
18	LC: 169. Majority Element	Boyer-Moore Majority Voting	Easy
19	LC: 11. Container With Most Water	Two Pointer	Med
20	LC: 75. Sort Colors	Dutch National Flag	Med
21	LC: 15. 3Sum	Two Pointers	Med
22	LC: 74. Search a 2D Matrix	Binary Search	Med
23	LC: 31. Next Permutation	Find rightmost ascending pair + swap + reverse	Med
24	LC: 56. Merge Intervals	Sorting and Greedy	Med
25	LC: 42. Trapping Rain Water	Two pointers	Hard
26	LC: 922. Sort Array By Parity II	Two Pointers	Easy
27	LC: 189. Rotate Array	Reverse 3 times	Med
28	LC: 2079. Watering Plants	Single pass simulation	Easy
29	LC: 2483. Minimum Penalty for a Shop	Prefix Sum	Med
30	LC: 442. Find All Duplicates in an Array	Marking index Negatively	Med
31	gfg: Kth smallest element		Med
32	LC: 238. Product of Array Except Self	Optimized prefix suffix	Med
33	gfg: Merge Overlapping intervals		Med

No.	Problem	Approach	Difficulty
34	LC: 54. Spiral Matrix	2D Matrix layer simulation	Med
35	LC: 59. Spiral Matrix II	2D Matrix layer simulation	Med
36	LC: 48. Rotate Image	2D Matrix: transpose + reverse rows	Med
37	LC: 867. Transpose Matrix	2D Matrix index swap	Easy
38	LC: 36. Valid Sudoku	2D Matrix: row/col/box hash sets	Med
39	LC: 794. Valid Tic-Tac-Toe State	2D Matrix + rule validation	Med
40	LC: 3195. Find the Minimum Area to Cover All Ones I	2D Matrix bounding box scan	Easy
Unit: I – Strings			
41	LC: 125. Valid Palindrome	Two pointers (alphanumeric check)	Easy
42	LC: 680. Valid Palindrome II	Two pointers with one skip	Easy
43	LC: 242. Valid Anagram	Frequency count (hash map)	Easy
44	LC: 14. Longest Common Prefix	Vertical scan / binary search	Easy
45	LC: 647. Palindromic Substrings	Expand around center	Med
46	LC: 44. Wildcard Matching	Dynamic programming / greedy	Hard
47	LC: 5. Longest Palindromic Substring	Expand around center / Manacher	Med
48	LC: 31. Next Permutation	Find rightmost ascending pair + swap + reverse	Med
Unit: I – Stack			
49	LC: 32. Longest Valid Parentheses	Stack tracking open parentheses indices	Hard
50	LC: 84. Largest Rectangle in Histogram	Monotonic stack (ascending)	Hard
51	LC: 85. Maximal Rectangle	Per-row histogram + monotonic stack	Hard
52	LC: 321. Create Maximum Number	Monotonic stack x2 + merge	Hard
53	LC: 678. Valid Parenthesis String	Greedy with low/high bounds	Med
54	LC: 735. Asteroid Collision	Stack simulation	Med
55	LC: 2751. Robot Collisions	Stack simulation	Hard
56	gfg: Implement two stacks in an array	Two-end array pointers	Easy
57	gfg: Sort a Stack using Recursion	Recursive insert in sorted order	Med
58	gfg: The Celebrity Problem	Two-pointer elimination	Med
59	gfg: Next Greater Element	Monotonic stack	Easy
60	LC: 3638. Maximum Balanced Shipments	Stack-based balancing	Hard
61	LC: 20. Valid Parentheses	Stack: push open, pop on close	Easy
Unit: II – Queue			
62	gfg: Rotten Oranges	BFS multi-source	Med
63	LC: 239. Sliding Window Maximum	Monotonic deque	Hard
64	LC: 622. Design Circular Queue	Array with head/tail pointers	Med
65	LC: 641. Design Circular Deque	Array with head/tail pointers	Med
66	LC: 2327. Number of People Aware of a Secret	Sliding window queue simulation	Med

No.	Problem	Approach	Difficulty
67	gfg: Iterative Tower of Hanoi	Stack-based iterative simulation	Med
Unit: III – Trees			
68	LC: 94. Binary Tree Inorder Traversal	Inorder DFS / Morris traversal	Easy
69	LC: 95. Unique Binary Search Trees II	Recursive backtracking	Med
70	LC: 98. Validate Binary Search Tree	DFS with min/max bounds	Med
71	LC: 101. Symmetric Tree	Mirror check DFS/BFS	Easy
72	LC: 102. Binary Tree Level Order Traversal	BFS level-by-level queue	Med
73	LC: 104. Maximum Depth of Binary Tree	DFS recursion	Easy
74	LC: 110. Balanced Binary Tree	Bottom-up DFS height check	Easy
75	LC: 113. Path Sum II	DFS backtracking with path	Med
76	LC: 124. Binary Tree Maximum Path Sum	DFS post-order with global max	Hard
77	LC: 199. Binary Tree Right Side View	BFS take last node per level	Med
78	LC: 297. Serialize and Deserialize Binary Tree	BFS/DFS serialize with delimiters	Hard
79	LC: 337. House Robber III	Tree DP (rob/not rob)	Med
80	LC: 450. Delete Node in a BST	BST in-order successor/predecessor	Med
81	LC: 449. Serialize and Deserialize BST	Preorder serialize + BST property	Med
82	LC: 617. Merge Two Binary Trees	DFS merge node values	Easy
83	LC: 814. Binary Tree Pruning	Post-order DFS pruning	Med
84	LC: 1123. Lowest Common Ancestor of Deepest Leaves	DFS: depth + LCA logic	Med
85	LC: 230. Kth Smallest Element in a BST	In-order traversal with counter	Med
86	LC: 106. Construct Binary Tree from Inorder and Postorder Traversal	Recursive: postorder gives root, inorder splits	Med
87	LC: 889. Construct Binary Tree from Preorder and Postorder Traversal	Recursive: pre/post give root & subtree split	Med
88	LC: 114. Flatten Binary Tree to Linked List	Morris traversal / DFS reverse	Med
Unit: III – Graphs			
89	LC: 79. Word Search	DFS + backtracking	Med
90	LC: 200. Number of Islands	DFS/BFS flood-fill	Med
91	LC: 207. Course Schedule	Topological sort (Kahn's / DFS cycle)	Med
92	LC: 210. Course Schedule II	Topological sort (Kahn's algo)	Med
93	LC: 365. Water and Jug Problem	BFS on state (x,y)	Med
94	LC: 124. Binary Tree Maximum Path Sum	DFS post-order with global max	Hard
95	LC: 127. Word Ladder	BFS word transformation	Hard
96	LC: 126. Word Ladder II	BFS + backtrack shortest paths	Hard
97	LC: 133. Clone Graph	DFS/BFS with hash map clone	Med
98	LC: 685. Redundant Connection II	Union-Find	Hard
99	LC: 913. Cat and Mouse	Minimax + BFS game theory	Hard
100	LC: 1192. Critical Connections in a Network	Tarjan's bridge finding algorithm	Hard

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101	LC: 1254. Number of Closed Islands	DFS boundary marking	Med
Unit: IV – Hashing			
102	LC: 12. Integer to Roman	Greedy with value-symbol table	Med
103	LC: 2295. Replace Elements in an Array	Hash map index remapping	Med
104	LC: 3120. Count the Number of Special Characters I	Set of lowercase & uppercase chars	Easy
105	LC: 3121. Count the Number of Special Characters II	Hash map last seen index tracking	Med
106	LC: 49. Group Anagrams	Hash map with sorted key	Med
107	LC: 229. Majority Element II	Boyer-Moore Majority Voting	Med
108	LC: 2350. Shortest Impossible Sequence of Rolls	Hash set of seen values	Hard
109	LC: 1452. People Whose Favorite Companies List Is Not Subset of other List	Set containment check per pair	Med
110	LC: 2963. Count the Number of Good Partitions	Hash map last occurrence tracking	Hard
111	LC: 2405. Optimal Partition of String	Greedy + sliding window hash set	Med
112	LC: 2584. Split the Array to Make Coprime Products	Hash map LCM tracking	Hard
113	LC: 299. Bulls and Cows	Hash map count (bulls & cows)	Med
114	LC: 822. Card Flipping Game	Hash set front/back tracking	Med
115	LC: 3185. Count Pairs That Form a Complete Day II	Hash map modular grouping	Med
116	LC: 2364. Count Number of Bad Pairs	Hash map counting pairs	Med
117	LC: 2763. Sum of Imbalance Numbers of All Subarrays	Hash map / sorted set per index	Hard
118	LC: 1001. Grid Illumination	Hash map diagonals + rows/cols	Hard
119	LC: 1224. Maximum Equal Frequency	Hash map frequency of frequencies	Hard
120	LC: 2561. Rearranging Fruits (imp)	Greedy + hash map frequency matching	Hard
Unit: V – Binary Search			
121	LC: 4. Median of Two Sorted Arrays	Binary search on smaller array partition	Hard
122	LC: 33. Search in Rotated Sorted Array	Binary search on rotated array	Med
123	LC: 34. Find First and Last Position of Element in Sorted Array	Two binary searches (lo/hi)	Med
124	LC: 69. Sqrt(x)	Binary search on answer	Easy
125	LC: 74. Search a 2D Matrix	Binary Search	Med
126	LC: 81. Search in Rotated Sorted Array II	Binary search (handle duplicates)	Med
127	LC: 153. Find Minimum in Rotated Sorted Array	Binary search on inflection point	Med
128	LC: 154. Find Minimum in Rotated Sorted Array II	Binary search (handle duplicates)	Hard
129	LC: 875. Koko Eating Bananas	Binary search on answers	Med
130	LC: 2226. Maximum Candies Allocated to K Children	Binary search on answers	Med
131	LC: 2187. Minimum Time to Complete Trips	Binary search on answers	Med
132	LC: 2448. Minimum Cost to Make Array Equal	Binary search on answers	Hard

No.	Problem	Approach	Difficulty
133	LC: 1760. Minimum Limit of Balls in a Bag	Binary search on answers	Med
134	LC: 2064. Minimized Maximum of Products Distributed to Any Store	Binary search on answers	Med
135	LC: 1552. Magnetic Force Between Two Balls	Binary search on answers	Med
136	gfg: Aggressive Cows	Binary search on answers	Med